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PRACTICAL 1: NETWORK TOOLS AND SAFETY

Objective

Identify networking tools and apply safe working procedures.

Activities

1. Identify networking tools.

- Crimping tool
- Punch down tool
- Cable tester
- Screw drivers

2. Explain the use of each tool.

Crimping tool

The crimping tool is used to crimp CAT cables with RJ-45 connectors by securing them to each other.

Punch down tool

The Punch down tool is used to place/push individual wires of a cable into the IDC slots on patch panels.

Cable tester

Is used to check the ordering of the wires on a network cable and it also shows if there is a fault in the cable by sending a signal through each wire in the networking cable.

Screw drivers

A screwdriver is a basic tool but is still used in networking, they are used to tighten or loosen any screws on network devices or they are used to mount the network devices to the wall like routers.

3. Demonstrate safe handling.

Crimping tool

- Make sure you have good grip on the crimping tool.
- Make sure you keep your fingers away from the blade at all times.

Punch down tool

- Do not press too hard on the wires to avoid damaging the wires.
- Use safety glasses.

Cable tester

- Do not connect it to live wires.
- Handle the cables with care to avoid damaging them.

Screwdrivers

- Use the correct size screwdriver to prevent it from slipping.
- Don't use broken screwdrivers.

4. Complete a tool inventory sheet.

Network Tools

Tools	Available	Condition
Toolbox	✓	Working
Lock	✓	Good
Ball of string	✓	Weak
Crossover Category 5e cable, 6 feet	✓	Good
Flashlight	✓	Good
Fish tape	✓	Good
Measuring tape	✓	Good
Needle nose pliers	✓	Weak
Network cable tester	✓	Working
Nut driver set	✓	Good
Punch down tool	✓	Good
RJ45 / RJ11 cable crimper	✓	Good
Safety goggles	X	N/A
Scissors	✓	Working
Screwdrivers	✓	Good
Soldering iron and accessories	✓	Good
Straight thru Category 5e cable, 6 feet	✓	Good
Stud finder	✓	Good
Two-way radio set	✓	Working
Utility knife	✓	Working
Wire cutters	✓	Weak
Wire strippers	✓	Good

Expendable Network Materials

Materials	Quantity
Assorted computer screws in a small container	100
Assorted RJ45 connectors	40
Batteries	1
Category 5e cable	100m
Electrical tape and duct tape	2 rolls
Nylon wire ties	100
Printed circuit board jumper	20
Rag / towels	5
Trash bag	10
Wire numbers set	1 set

5. Apply health and safety procedures.

- Read the appropriate manufacturers instruction .
- Identify potential hazards.
 - Loose cables
 - Sharp Tools
 - Nearby electrical equipment.
- Wear the appropriate personal protective equipment.
 - Safety goggles
 - Protective gloves
 - Safety boots
 - Dust mask
- Keep your work area clean and organized.
- Don't use faulty equipment.

Deliverable		
Completed Tool Identification Worksheet.		
Assessment Rubric (10 Marks) Criteria	Marks	
Tool identification	4	
Understanding tool functions	2	
Safe tool usage	2	
Worksheet completion	2	
Total	10	